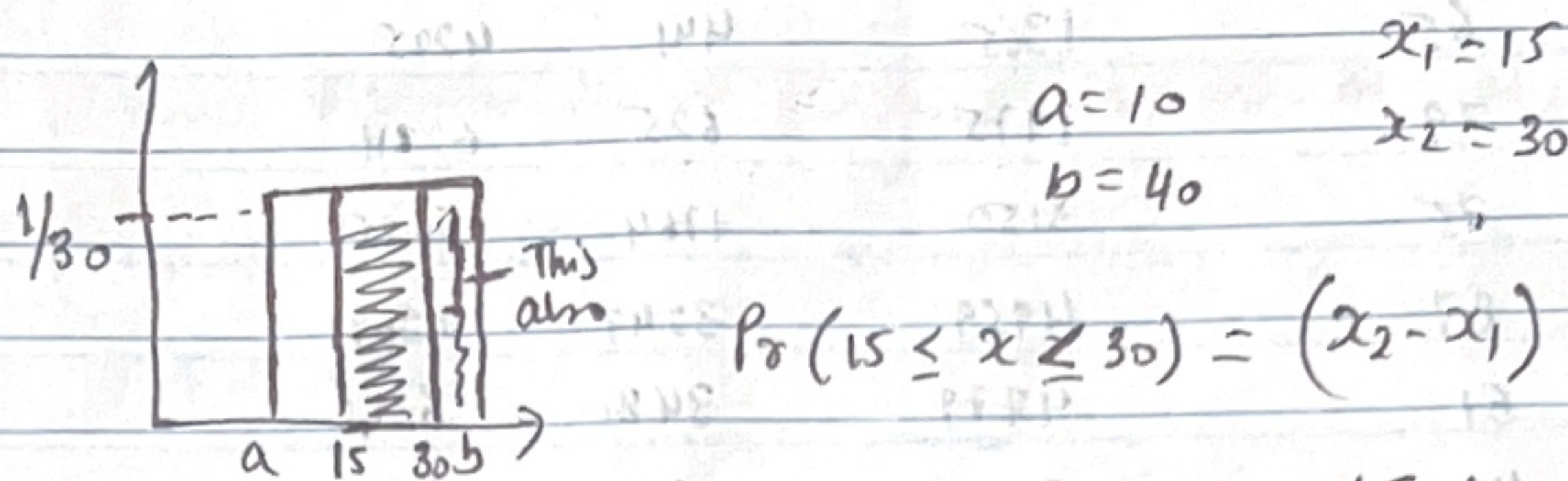


Q. The # of candies sold daily at a shop is uniformly distributed with a maximum of 40 and min of 10

Probability of daily sales to fall between 15 and 30?



$$Pr(15 \leq x \leq 30) = (x_2 - x_1) * \frac{1}{b-a}$$

$$= 15 * \frac{1}{30-10} \Rightarrow 0.5$$

$$Pr(x > 30) = (40 - 30) * \frac{1}{30} \Rightarrow \frac{1}{3} = 0.33$$

Linear Regression with one Independent Variable

Independent Variable

Dependent Variable

Grades level (4)

Subject

Age (x)

99

1

43

2

21

65

3

25

79

4

42

75

5

57

87

6

59

81

7

55

?

$$y_i = b_0 + b_1 x_i$$

Estimated y value

Intercept

Value of x for observation i

$$b_0 = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b_1 = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

Subject	Age (x)	Glucose level (y)	x^2	xy	y^2
1	43	99	1849	4257	9801
2	21	65	441	1365	4225
3	25	79	625	1975	6241
4	42	75	1764	3150	5625
5	57	87	3249	4959	7569
6	59	81	3481	4779	6561
Σ	247	486	11409	20485	40022

$$b_0 \Rightarrow \frac{(486)(11409) - (247)(20485)}{6(11409) - (247)^2} = \frac{4848979}{7445} = 65.14$$

$$b_1 \Rightarrow \frac{6(20485) - (247)(486)}{6(11409) - (247)^2} = \frac{2868}{7445} = 0.385335$$

$$y' = b_0 + b_1 x$$

$$y' = 65.14 + (0.385335)x$$

Given $x = 55$

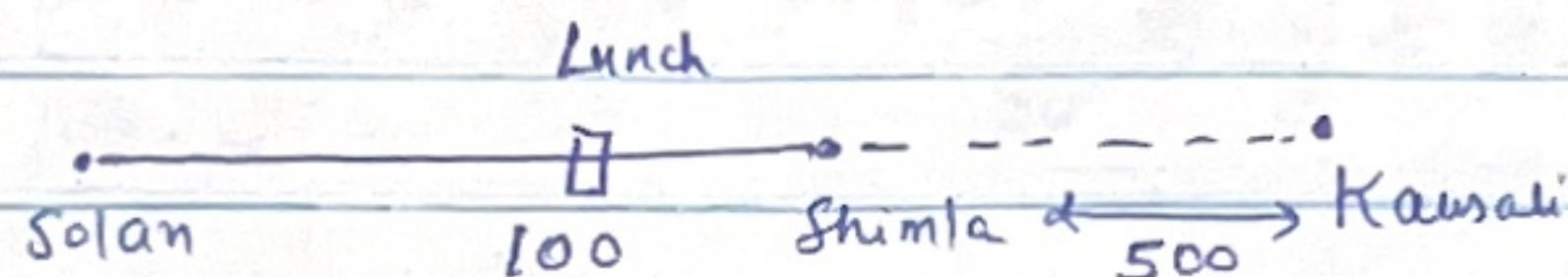
$$\begin{aligned} f(x) &= 65.14 + (0.385335 \times 55) \\ &= 86.327 \end{aligned}$$

Find the intercept of linear Regression line if $\sum x = 25$, $\sum y = 20$, $\sum x^2 = 90$, $\sum xy = 150$, $n = 5$

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2} \Rightarrow \frac{20(90) - 25(150)}{5(90) - 625} = \frac{-1950}{-175} \Rightarrow 11.14$$

Find the slope of linear Regression line if $\sum x = 10$, $\sum y = 16$, $\sum x^2 = 60$, $\sum xy = 120$ and $n = 4$

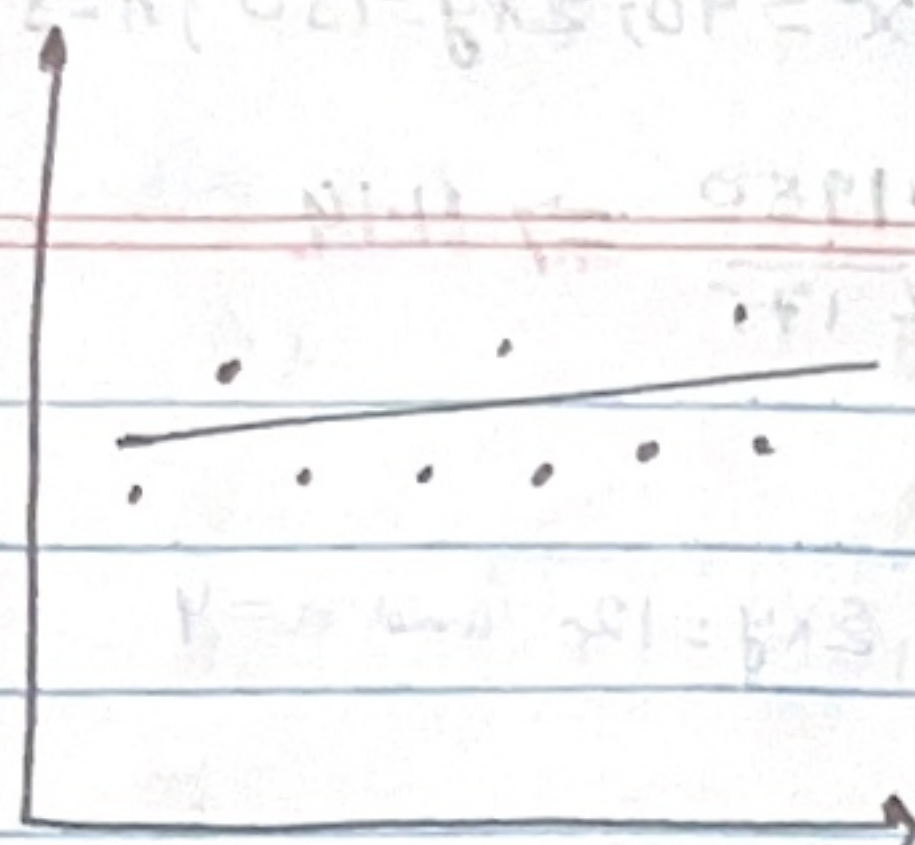
$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2} = \frac{4(120) - 10(16)}{4(60) - 100} = \frac{320}{140} = 2.29$$



2000Rs $\Rightarrow$ Solan to Shimla

$$\begin{aligned} y &= 2000 + 100 * 500 \\ &= 2000 + 50000 \\ &= 52,000 \end{aligned}$$

Least Square Method $\Rightarrow$ determine line of best fit



Reduce the sum of the squares of vertical deviation from each data point to line

For any line fitted accurately its perpendicular deviation is zero.

Error $\rightarrow$ Standard Error

$$P.L.M = \frac{\sum y}{\sum 1} = \frac{(\sum y) - (\sum x)(\sum y)}{\sum 1 - (\sum x)^2} = \frac{(\sum y) - (\sum x)(\sum y)}{\sum 1 - (\sum x)^2}$$